

Problem-Solution Fit

Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The Problem-Solution Fit Canvas is a strategic template designed to align a target audience's core problems, behaviors, and limitations directly with a proposed solution. For the RISING WATERS framework, it maps the severe gaps in conventional flood forecasting against an advanced, high-accuracy machine learning system and accessible web dashboard.

1. CUSTOMER SEGMENT(S) [CS] 1. Residents living in high-risk, flood-prone districts. 2. Regional meteorologists and environmental analysts. 3. Disaster relief coordinators and response teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] 1. Minimal budget to implement private, dedicated sensor networks. 2. Limited to basic consumer devices (smartphones/web browsers).	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: Macro weather reports give general regional overviews. Cons: Outdated, manual tracking fails to provide timely micro warnings.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] 1. Problem: Devastating annual loss of local lives and household assets. 2. Frequency: Recurring predictably every monsoon season. 3. Pain: Receiving late, highly insufficient advance warnings.	9. PROBLEM ROOT / CAUSE [RC] 1. Existing prediction systems are not accessible to the general public. 2. Failure of traditional tools to process complex weather patterns quickly.	7. BEHAVIOR + ITS INTENSITY [BE] 1. High Intensity: Paralyzing fear and vulnerability during monsoons. 2. Chaotic, last-minute scrambles to relocate assets safely.

3. TRIGGERS TO ACT [TR]

1. Sudden onset of severe, localized rainfall patterns.
2. Rapidly dropping cloud visibility levels during monsoons.
3. Surging regional water logging and alarming weather metrics.

4. EMOTIONS BEFORE / AFTER [EM]

- 1.Before:** Extremely anxious, stressed, and completely unprepared.
- 2.After:** Reassured, confident, and proactive with early risk classifications.

10. YOUR SOLUTION [SL]

- 1. Project Name:** RISING WATERS.
2. High-accuracy (96.55%) XGBoost machine learning predictive model.
3. Intuitive, interactive Flask-based early warning web dashboard.
4. Scalable deployment on IBM Cloud for instant multi-device access.

8. CHANNELS OF BEHAVIOR [CH] ONLINE

- 1.Weather Portals:** Manually browsing open-source repositories and online platforms like Kaggle to track general meteorological parameters and historical data patterns.
- 2.System Dashboards:** Accessing digital web application frameworks via browsers on consumer devices to look for localized, real-time predictive hazard reports.

OFFLINE

- 1.Visual Ground Observations:** Manually monitoring visible changes in local cloud patterns, rain density, and nearby water logging levels.
- 2. Community Communication:** Relying on basic word-of-mouth alerts, local sirens, or physical emergency instructions to coordinate immediate last-minute relocations.